

# Reverse BAA / Program Challenge Statement

Governed Execution Substrates for High-Consequence Autonomy

***No proof, no actuation.***

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## Heilmeier Catechism

### What are you trying to do?

Build a runtime layer that prevents high-consequence autonomous systems from producing real-world effects unless they can prove valid human authority, provenance, restraint, and continuity of command at the exact moment of action. *If the system cannot prove it is still under valid human command, it cannot act.*

### How is it done today?

Autonomy is governed through design-time approval, test and evaluation, assurance cases, doctrine, policy, operator training, logging, and after-action review.

### What is the limit of that approach?

Those methods can show a system was approved before deployment. They do not prove that a specific machine-speed action remains under valid human command at the moment it occurs.

### What is new?

This proposal treats human command as a runtime-verifiable execution condition, not a legal or doctrinal claim. The core rule: *No proof, no actuation*. An autonomous system must carry or obtain machine-checkable proof of authority, provenance, restraint, and command continuity before it can create a high-consequence effect. The verification primitive is intentionally left open — signed capability tokens, zero-knowledge proofs of authorization, Merkle-chained command logs — because selecting among them is part of what this program should force.

### Why might it work?

The problem reduces to enforceable runtime predicates: authority binding, tamper-evident provenance, constraint verification, revocation handling, and fail-closed execution gates. Each is independently testable and blockable. No action crosses the actuation boundary unless all predicates pass.

### Who cares?

Anyone responsible for fielding autonomous systems where speed, degraded communications, adversarial manipulation, or mission drift could cause action to occur faster than meaningful human review.

### What difference would success make?

"Meaningful human command" becomes a testable system property, not a policy assertion.

### What are the risks?

Latency, false blocks, degraded-comms failures, and bypass by poorly integrated platforms. These are the technical exams the program should force. The degraded-comms case is not solved by a blanket fail-closed rule — bounded-time local verification is an explicit research subproblem: what is the minimum authority token a platform may carry offline, for how long, and under what revocation constraints?

### What are the exams?

A governed autonomous platform must pass these tests:

- (1) Valid command permits action.

- (2) Revoked command blocks action.
- (3) Broken provenance blocks action.
- (4) Constraint violation blocks or modifies action.
- (5) Model update does not erase command accountability.
- (6) Degraded communications do not silently convert autonomy into uncommanded action.
- (7) Attempted override is detected, logged, and blocked.
- (8) Runtime overhead remains within operational limits.

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## **The Gap This Program Fills**

Existing DARPA programs address whether a system was safe before deployment. No current portfolio effort addresses whether a specific action is authorized at the moment it fires. When systems act at machine speed — after model updates, under degraded comms, against adversarial context injection — design-time assurance cannot reach the actuation boundary. The missing layer is a runtime substrate that treats authority, provenance, restraint, and continuity of command as hard blocking predicates, not post-hoc audit trails. This is a distinct, unsolved architectural problem.

## **Demonstration**

A simulated or hardware-in-the-loop platform — UAV, UGV, maritime system, or decision-support node — in which actuator-level actions are structurally impossible unless the substrate verifies all four predicates at the moment of contemplated action. Required test cases: valid command, revoked command, degraded provenance, attempted adversarial override, and a comms-denied scenario in which the platform must rely solely on a locally-held authority token.

## **Routing Request**

The author requests that the relevant program manager — ATO, I2O, or equivalent — identify the appropriate pathway for a short concept paper further developing this program challenge. ERIS topic area or direct BAA pathway both acceptable.

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This document is a reverse BAA — a program challenge statement identifying a missing execution-governance problem the author believes DARPA is structurally positioned to formalize. It is not a formal proposal, white paper, or response to an existing solicitation.

Jeremy Karrick — Principal Architect, Thirsty's Projects | [Karrick1995@gmail.com](mailto:Karrick1995@gmail.com) | 385-522-9743 | [www.thirstysprojects.com](http://www.thirstysprojects.com)